

Cyclically 4-Edge-Connected Cubic Graphs of Unbounded Petersen Defect

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CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

The Petersen defect of a cubic graph is the least number of vertices at which an edge map into the Petersen graph fails to send the vertex star onto a vertex star; it is zero exactly for the graphs with a Petersen coloring, and it bounds from below the least number of abnormal edges of a proper 5-edge-coloring. Mattiolo, Mazzuoccolo and Mkrtchyan conjectured that five statements are equivalent: the Petersen coloring conjecture, and the existence of a sublinear bound on the number of abnormal edges for all bridgeless, all 2-connected, all 3-connected, and all cyclically 4-edge-connected cubic graphs. After the refutation of the Petersen coloring conjecture in 2026, the first two statements are false, ring and frame constructions show that the third and fourth are false too, and the conjecture reduces to the last one. We show that it is false as well: for every $t \geq 2$ there is a cyclically 4-edge-connected cubic graph of girth 5 on $102t$ vertices whose Petersen defect is at least t , so every proper 5-edge-coloring of it has at least t abnormal edges. Hence all five statements are false and the conjectured equivalence holds. The graphs are rings that alternate two 4-poles cut from a 52-vertex counterexample of Goedgebeur, Jookan, Máčajová, Mattiolo, Mazzuoccolo and Ulyanov. The argument rests on a conservation law: modulo the cut space of the Petersen graph, a 4-pole whose vertices are all correctly mapped transmits the same class through both of its connectors, and the class of a pair of labels is their distance in the line graph of the Petersen graph. One 4-pole transmits only distance 1, the other only distance 0, so no two consecutive blocks of a ring can both be free of bad vertices. The two transmitted sets and the non-colorability of the smallest ring are certified by SAT refutations with DRAT proofs checked by **drat-trim**, and the cyclic 4-edge-connectivity of all rings follows from that of the four-block ring, verified by two independent exhaustive searches. The ring on 306 vertices has Petersen defect exactly 3. For rings joined along 3-edge connectors, the class of a vertex modulo the cut space is one of 64 cosets of the cut code of the Petersen graph, in six orbits under its automorphism group, and the transfer relation of a 6-pole is block diagonal over these classes; so the Petersen colorability of every ring built from a finite family of 6-poles, of every length, is decided by a finite semigroup computation with a re-checkable certificate. Every ring of claws, of Petersen superedges, or of both, is Petersen colorable.

This covers the flower snarks, whose junctions always receive three pairwise antipodal edges of the Petersen graph, and an infinite family of cyclically 5-edge-connected snarks of girth 5. Splitting a 52-vertex counterexample along a 6-edge cut gives two colorable 6-poles that conduct the same classes, so the classes alone do not detect its non-colorability, while the transfer relation does.

1 Introduction

Let P be the Petersen graph. A *Petersen coloring* of a cubic graph G is a map $\sigma : E(G) \rightarrow E(P)$ such that for every vertex v of G the three edges at v are sent bijectively onto the three edges at some vertex of P . Jaeger conjectured that every bridgeless cubic graph has one [1], and showed that a cubic graph has a Petersen coloring if and only if it has a *normal* proper 5-edge-coloring, one in which every edge is poor (its two end stars together see three colors) or rich (they see five) [2]. The conjecture was refuted in August 2026 by Putman [3], with a human-checkable proof by Jookan [4]; Goedgebeur, Jookan, Máčajová, Mattiolo, Mazzuoccolo and Ulyanov [5] found two counterexamples on 52 vertices and cyclically 4-edge-connected counterexamples of every even order at least 60.

An edge of a proper 5-edge-coloring that is neither poor nor rich is *abnormal*; write $\text{ab}(G)$ for the least number of abnormal edges over all proper 5-edge-colorings of G . Mattiolo, Mazzuoccolo and Mkrtchyan [9] proved that no proper 5-edge-coloring has exactly one abnormal edge and made the following conjecture.

Conjecture 1.1 ([9], Conjecture 3). *The following statements are equivalent: (a) the Petersen coloring conjecture holds; there is a sublinear function f with $\text{ab}(G) \leq f(|V(G)|)$ for (b) all bridgeless, (c) all 2-connected, (d) all 3-connected, respectively (e) all cyclically 4-edge-connected cubic graphs G .*

They proved that (a) and (b) are equivalent, and that on each of the last three classes a sublinear bound exists if and only if ab is bounded by 5, 7 and 9 respectively [9, Theorems 1–4]. After the disproof, (a) and (b) are false. Rings of counterexamples joined through 2-edge cuts, and cubic frames whose vertices are replaced by counterexamples minus a vertex, have one bad vertex in every block, so (c) and (d) are false as well [10]. The same work shows that, given a proposition of [9] on cyclic joins, (e) is equivalent to every cyclically 4-edge-connected cubic graph admitting a map into $E(P)$ with at most two bad vertices, and it records that in fifteen cyclically 4-edge-connected counterexamples every pair of adjacent vertices is critical. The conjecture therefore turned on statement (e). We settle it; the proof does not use that proposition.

Theorem 1.2. *For every $t \geq 2$ there is a cyclically 4-edge-connected cubic graph R_t of girth 5 on $102t$ vertices such that every map $E(R_t) \rightarrow E(P)$ has at least t vertices at which the star condition fails. Consequently every proper 5-edge-coloring of R_t has at least t abnormal edges.*

Corollary 1.3. *No sublinear function bounds the number of abnormal edges on cyclically 4-edge-connected cubic graphs. Hence all five statements of Conjecture 1.1 are false, and the conjecture holds.*

The graphs R_t are rings of $2t$ blocks that alternate two 4-poles, A and B , both cut from the first 52-vertex counterexample G_{52} of [5]: A is G_{52} with two independent edges removed, and B is G_{52} with two adjacent vertices removed. The proof uses a conservation law. Modulo the

cut space of P , a 4-pole whose vertices all satisfy the star condition transmits the same class through its two connectors (Lemma 3.3), and the class carried by a pair of edges is the distance of their labels in the line graph of P (Lemma 3.2). The 4-pole A transmits only distance 1 and B only distance 0 (Proposition 5.1), so two consecutive blocks of a ring cannot both be free of bad vertices (Theorem 4.1). Two small restoration arguments (Lemma 4.2) show that each of the two computed restrictions is enough on its own. The rings are cyclically 4-edge-connected because the four-block ring is (Lemma 4.3).

The rings have 4-edge cuts, and every known counterexample to the Petersen coloring conjecture does; whether a cyclically 5-edge-connected one exists is Problem 11 of [5]. Section 6 examines the natural analogue of the construction for that class, rings of 6-poles joined along 3-edge connectors. The class of a connector modulo the cut space is still conserved, but on the 6-poles of the Petersen graph, the flower snarks J_5 , J_7 and the dodecahedron it never separates two blocks. The exact transfer relation is block diagonal over the 64 classes, and the colorability of every ring of a finite family of blocks reduces to a finite semigroup. For the claw and the Petersen superedge every ring is colorable (Theorem 6.4).

2 Preliminaries

Graphs are finite. For a vertex v , $\partial(v)$ is the set of edges at v ; for a vertex set S , $\delta(S)$ is the set of edges with exactly one end in S . A cubic graph is *cyclically 4-edge-connected* if every edge cut whose two sides both contain a cycle has at least four edges.

Maps into the Petersen graph. For a cubic graph G and a map $\sigma : E(G) \rightarrow E(P)$, a vertex v is *good* if σ maps $\partial(v)$ bijectively onto the star of a vertex of P , and *bad* otherwise. The *Petersen defect* $\text{pd}(G)$ is the least number of bad vertices over all maps; $\text{pd}(G) = 0$ exactly when G has a Petersen coloring. For v let $\chi_v \in \mathbb{F}_2^{E(P)}$ be the sum of the indicator vectors of the three labels at v , and let $\text{Cut}(P)$ be the cut space of P . The star of a vertex is a cut, so $\chi_v \in \text{Cut}(P)$ when v is good; for every map the sum of χ_v over all vertices is zero, because every edge is counted at both ends. The parity theorem of [10] follows: the number of bad vertices is never exactly one, so every graph without a Petersen coloring has $\text{pd} \geq 2$.

Abnormal edges. We use the Kneser model of P : its vertices are the 2-subsets of $\{1, \dots, 5\}$, two of them adjacent when disjoint, and the edge $\{X, Y\}$ has the color that lies outside $X \cup Y$, so the three edges at X carry the three colors outside X .

Lemma 2.1 ([10]). $\text{pd}(G) \leq \text{ab}(G)$ for every cubic graph G .

Proof. Given a proper 5-edge-coloring c , let X_v be the pair of colors missing at v , and let the *view* of an edge $e = uv$ from u be the edge of P at X_u of color $c(e)$. If e is poor then $X_u = X_v$; if it is rich then $X_u \cap X_v = \emptyset$ and both avoid $c(e)$; in both cases the two views coincide. Map every normal edge to its common view and every abnormal edge to its view from one chosen end. A vertex all of whose edges carry its own views is good, so the bad vertices are among the ends not chosen of the abnormal edges. $\square$

Multipoles. A 4-pole is obtained from a cubic graph by removing some vertices and edges so that exactly four *dangling ends* remain: an edge removed with one end kept, or kept at one end only, becomes a dangling end at the kept vertex. The dangling ends are grouped into two

connectors of two ends each. A map of a 4-pole assigns a label to every edge and to every dangling end; the star condition is required at the vertices of the 4-pole.

3 Classes of pairs of labels

The line graph of P has diameter 3: from a fixed edge, four edges are at distance 1, eight at distance 2 and two at distance 3. We write $d(x, y)$ for this distance.

Lemma 3.1. *The cuts of P with at most four edges are the empty cut, the ten stars, and the fifteen sets of four edges adjacent to one edge.*

Proof. P is 3-edge-connected, cyclically 5-edge-connected and of girth 5. If one side S of a cut induces a forest with k components, the cut has $3|S| - 2(|S| - k) = |S| + 2k$ edges, which is 3 only for a single vertex and 4 only for two adjacent vertices; if both sides contain a cycle, the cut has at least five edges. $\square$

Lemma 3.2. *Let x, y, x', y' be edges of P . If $\mathbf{1}_x + \mathbf{1}_y$ and $\mathbf{1}_{x'} + \mathbf{1}_{y'}$ lie in the same class modulo $\text{Cut}(P)$, then $d(x, y) = d(x', y')$.*

Proof. The class of $\mathbf{1}_x + \mathbf{1}_y$ is zero exactly when $x = y$, since a vector of weight 2 is not a cut. If $x \neq y$, $x' \neq y'$ and the two sums have the same class, the symmetric difference of $\{x, y\}$ and $\{x', y'\}$ is a cut of even weight at most 4. By Lemma 3.1 it is empty, so that the pairs coincide, or it consists of the four edges adjacent to an edge st of P . Then either each pair consists of two edges at the same end of st , and both pairs are at distance 1, or each pair has one edge at s and one at t , and both pairs are at distance 2 because P has no triangle. $\square$

Lemma 3.3 (conduction). *Let M be a 4-pole with connectors $\{p, q\}$ and $\{r, s\}$, and let σ be a map of M in which every vertex of M is good. Then $d(\sigma(p), \sigma(q)) = d(\sigma(r), \sigma(s))$.*

Proof. The sum of χ_v over the vertices of M is a sum of stars, hence a cut. An edge of M is counted at both of its ends, so the sum equals $\mathbf{1}_{\sigma(p)} + \mathbf{1}_{\sigma(q)} + \mathbf{1}_{\sigma(r)} + \mathbf{1}_{\sigma(s)}$. The two connectors therefore carry the same class, and Lemma 3.2 gives the claim. $\square$

For a 4-pole M with its connectors, the *transmitted set* $D(M) \subseteq \{0, 1, 2, 3\}$ is the set of distances $d(\sigma(p), \sigma(q))$ over all maps σ of M with every vertex good. By Lemma 3.3 it does not depend on which connector is read. Because the automorphism group of P is transitive on ordered pairs of edges at a given distance, $d \in D(M)$ can be decided by one formula in which the labels of p and q are fixed to one such pair.

4 Rings of 4-poles

Let A and B be 4-poles, each with a first and a second connector. The ring $R_t(A, B)$ consists of t copies of A and t copies of B placed alternately around a cycle, the second connector of each block joined by two edges to the first connector of the next block. It is cubic. The two edges that join consecutive blocks form a connector of both blocks.

Theorem 4.1. *If $D(A) \cap D(B) = \emptyset$, then every map of $R_t(A, B)$ has bad vertices in at least t of its $2t$ blocks. In particular $\text{ab}(R_t(A, B)) \geq \text{pd}(R_t(A, B)) \geq t$.*

Proof. If two consecutive blocks had only good vertices, Lemma 3.3 applied to each of them would place the distance of their common connector in $D(A) \cap D(B)$. So the blocks without bad vertices are pairwise non-consecutive, and at most t of the $2t$ blocks are among them. The last inequality is Lemma 2.1. $\square$

Lemma 4.2 (restoration). *Let G be a cubic graph without a Petersen coloring.*

- (a) *If ab and cd are independent edges and $M = G - ab - cd$ has connectors $\{a, b\}$, $\{c, d\}$ (the dangling ends at these vertices), then $0 \notin D(M)$.*
- (b) *If uv is an edge, u_1, u_2 are the other neighbors of u and v_1, v_2 those of v , and $M = G - \{u, v\}$ has connectors $\{u_1, u_2\}$, $\{v_1, v_2\}$, then $1 \notin D(M)$.*

Proof. (a) Distance 0 means equal labels on a and b , and then by Lemma 3.3 equal labels on c and d ; putting these labels on ab and cd gives a Petersen coloring of G . (b) If the labels at u_1, u_2 are at distance 1, they share a vertex of P whose third edge z satisfies $\mathbf{1}_{\sigma(uu_1)} + \mathbf{1}_{\sigma(uu_2)} \equiv \mathbf{1}_z$ modulo $\text{Cut}(P)$; likewise the labels at v_1, v_2 give an edge z' . By Lemma 3.3 the two classes agree, so $\mathbf{1}_z + \mathbf{1}_{z'}$ is a cut and $z = z'$. Labelling uv by z makes u and v good, which gives a Petersen coloring of G . $\square$

So for $A = G - ab - cd$ and $B = G' - \{u, v\}$ with G, G' without Petersen colorings, the sets $D(A)$ and $D(B)$ are disjoint as soon as $D(A) \subseteq \{1\}$ or $D(B) \subseteq \{0\}$.

Lemma 4.3. *Let G be a cyclically 4-edge-connected cubic graph, $A = G - ab - cd$ and $B = G - \{u, v\}$ as in Lemma 4.2, with A joined through its connector $\{c, d\}$ to the connector $\{u_1, u_2\}$ of the next B , and B through $\{v_1, v_2\}$ to the connector $\{a, b\}$ of the next A . If $R_2(A, B)$ has no cycle-separating edge cut with at most three edges, then every $R_t(A, B)$ with $t \geq 2$ is cyclically 4-edge-connected.*

Proof. G is 3-edge-connected, so A is connected; G is 3-connected, so B is connected. Add to a block a vertex s^* joined to the two ends of its first connector and a vertex t^* joined to the two ends of its second connector. For B this gives $G - uv$ with u, v renamed s^*, t^* , which has two edge-disjoint s^*-t^* paths because G is 3-edge-connected. For A , an edge separating s^* from t^* would either separate $\{a, b\}$ from $\{c, d\}$ in G by itself, contradicting 3-edge-connectivity, or be an edge s^*a (or s^*b, t^*c, t^*d) whose removal leaves b (respectively a, d, c) without a path to the other connector, contradicting the connectivity of A . Consequently a ring carries four edge-disjoint paths between any two of its blocks, two along each arc.

Let K be a cycle-separating cut of R_t with at most three edges and sides S, S' . By the four paths, no block lies entirely in S while another lies entirely in S' ; say that every block not split by K lies in S . Every split block contains an edge of K , being connected, so at most three blocks are split and S' lies within them. A cycle in S' lies in a maximal run Y of consecutive split blocks; Y has at most three blocks and is bounded on both sides by blocks in S . The edges leaving $S' \cap V(Y)$ belong to K and form a cycle-separating cut whose size depends only on Y and $S' \cap V(Y)$. The ring $R_2 = ABAB$ contains every alternating run of at most three blocks with at least one further block around it, so it contains the same cut, contradicting the hypothesis. $\square$

5 The construction

Let G_{52} be the 52-vertex counterexample of [5] (House of Graphs entry 57244), with its vertices numbered $0, \dots, 51$ as in the edge list of the repository (SHA-256 of the sorted edge list 27db5d3b...); it is cyclically 4-edge-connected with girth 5 and has no Petersen coloring [5, 10]. Let

$$A = G_{52} - \{0, 3\} - \{1, 9\}, \quad B = G_{52} - \{2, 7\},$$

where $\{0, 3\}$, $\{1, 9\}$ are independent edges and $\{2, 7\}$ is an edge, with connectors as in Lemma 4.2: $\{0, 3\}$ and $\{1, 9\}$ for A , the two other neighbors of 2 and the two other neighbors of 7 for B . Let $R_t = R_t(A, B)$, joined as in Lemma 4.3; it has $52t + 50t = 102t$ vertices.

Proposition 5.1 (machine-verified). $D(A) = \{1\}$ and $D(B) = \{0\}$.

Proof. For each distance d one propositional formula states that the 4-pole has a map with every vertex good and the first connector labelled by a fixed pair at distance d . The satisfiable formulas give maps that an independent checker accepts; the others are refuted with DRAT proofs accepted by `drat-trim` (Table 1). Distance 0 for A and distance 1 for B are also excluded by Lemma 4.2. $\square$

4-pole	$d = 0$	$d = 1$	$d = 2$	$d = 3$
$A = G_{52} - \{0, 3\} - \{1, 9\}$	refuted (7.7 s)	map (0.5 s)	refuted (2.7 s)	refuted (2.1 s)
$B = G_{52} - \{2, 7\}$	map (0.2 s)	refuted (2.8 s)	refuted (2.2 s)	refuted (2.1 s)

Table 1: Transmitted distances of the two 4-poles. “Map”: a map with every vertex good and the first connector at distance d , accepted by the checker (the second connector is then at the same distance). “Refuted”: unsatisfiable, with a DRAT proof accepted by `drat-trim`. Times are solver times.

Proposition 5.2 (machine-verified). *The graphs R_1 and R_2 are simple cubic graphs of girth 5 without cycle-separating edge cuts of at most three edges, and the four edges joining one block to the rest form a cycle-separating cut, so their cyclic edge connectivity is exactly 4. R_1 has no Petersen coloring, and $\text{pd}(R_2) = 2$.*

Proof. The cut searches are exhaustive: for every pair of edges, the bridges of the graph without that pair are computed, and every edge set of size at most three found in this way is tested for leaving two sides that contain cycles; the same routine applied to G_{52} , to a ring of two copies of G_{52} opened at one edge, and to the cubic graph K_4 with its vertices replaced by copies of G_{52} minus a vertex returns no cut, 2-edge cuts, and 3-edge cuts respectively, in agreement with an enumeration of all edge sets of size at most three. That enumeration, run on R_1 and R_2 themselves, finds no cycle-separating cut either. The Petersen coloring formula of R_1 is refuted with a DRAT proof accepted by `drat-trim`. For R_2 , a map with exactly two bad vertices, both in the two copies of B , is accepted by the checker; two is the least possible by the parity theorem. $\square$

Proof of Theorem 1.2. By Proposition 5.1, $D(A) \cap D(B) = \emptyset$, so Theorem 4.1 gives $\text{ab}(R_t) \geq \text{pd}(R_t) \geq t$. By Proposition 5.2 and Lemma 4.3, R_t is cyclically 4-edge-connected for every $t \geq 2$. A cycle of length at most four would lie within two consecutive blocks and their joining edges, so it would occur in R_2 , which has girth 5. $\square$

Proposition 5.3 (machine-verified). $\text{pd}(R_3) = 3$.

Proof. The lower bound is Theorem 4.1; it was also tested directly on the 306 vertices of R_3 : the formula stating that some map has at most two bad vertices is unsatisfiable, with a DRAT proof accepted by `drat-trim` (5,802 seconds of solving and 2,265 seconds of checking). For the upper bound, the star condition is relaxed at one vertex in each of the three copies of B , the vertex at the position (within its copy) of one of the two bad vertices of a two-defect map of R_2 ; the relaxed formula is satisfiable (169 seconds), and the checker finds exactly these three vertices bad. $\square$

Proof of Corollary 1.3. The graphs R_t have $n = 102t$ vertices and $\text{ab}(R_t) \geq n/102$, so no sublinear function bounds ab on cyclically 4-edge-connected cubic graphs, and (e) is false; for $t \geq 10$ they also violate the bound 9 of [9, Theorem 4]. Statements (a) to (d) are false, as recalled in the introduction, so the five statements are equivalent. $\square$

6 6-poles: classes and transfer relations

The 64 classes. Write $Q = \mathbb{F}_2^{E(P)} / \text{Cut}(P)$, and call the class in Q of the sum of the indicator vectors of a multiset of labels the class of that multiset. The cut space is a binary [15, 9, 3] code (its nonzero words of weight at most 4 are those of Lemma 3.1), so Q has 64 elements. A vertex is good exactly when the class of its three labels is 0 and they are distinct. The automorphism group of P acts on Q with six orbits (Table 2); we call them 0, E, D_3 , D_2 , T_1 , T_2 . The orbit T_1 is the class of the all-ones vector: removing from $E(P)$ the twelve edges at the four 2-subsets containing a fixed element of $\{1, \dots, 5\}$ leaves three pairwise antipodal edges (at distance 3 in the line graph), and these five *antipodal triples* partition $E(P)$.

orbit	classes	least weight	elements of least weight
0	1	0	the cut space, containing the ten stars
E	15	1	one edge; also two edges at distance 1
D_3	15	2	two edges at distance 3
D_2	30	2	two edges at distance 2 (two such pairs per class)
T_1	1	3	the five antipodal triples
T_2	2	3	triples of pairwise distance-2 edges (ten per class)

Table 2: The orbits of $\text{Aut}(P)$ on the 64 classes modulo the cut space (computed and checked against Lemma 3.2).

6-poles. A *6-pole* has six dangling ends, split into two connectors of three. The proof of Lemma 3.3 shows that a map with every vertex good puts the same class on both connectors, and the *conducted set* $D(M) \subseteq Q$ is the set of these classes; it is invariant under $\text{Aut}(P)$, hence a union of orbits. Theorem 4.1 holds verbatim for rings of 6-poles joined along their connectors by three edges, and such rings can be cyclically 5-edge-connected, since two arcs of a ring are separated by six edges. For a graph G without a Petersen coloring and non-adjacent vertices u, w , the 6-pole $G - u - w$ with the ends at u and at w as connectors does not conduct 0, by the argument of Lemma 4.2: class 0 at the ends of u means three labels forming a star.

Proposition 6.1 (machine-verified). *Let H be the Petersen graph, the flower snark J_5 or J_7 , or the dodecahedron. Every 6-pole $H - u - w$ (u, w non-adjacent, connectors at u and at w), and, for $H \neq J_7$, every 6-pole $H - e_1 - e_2 - e_3$ (e_1, e_2, e_3 pairwise disjoint, any split of the six ends) conducts 0, E and D_2 ; the set $\{0, E, D_2\}$ itself is conducted by $P - u - w$.*

Proof. One formula for each 6-pole up to automorphisms of H , each split and each orbit, with the class of the first connector fixed to a representative of the orbit (1,890 splits, 11,340 formulas). The satisfiable ones give maps accepted by the checker, the others are refuted with DRAT proofs accepted by **drat-trim**; none of the four graphs has a cycle-separating cut of at most four edges. $\square$

So no two of these 6-poles have disjoint conducted sets, and Theorem 4.1 cannot show that a ring of them has no Petersen coloring. The exact obstruction is the following relation.

Transfer relations. A *block* is a 6-pole whose connectors L and R are ordered. Its transfer relation is the set T_M of pairs $(\sigma(L), \sigma(R)) \in E(P)^3 \times E(P)^3$ over all maps σ with every vertex good. In a ring of blocks $M_1, \dots, M_t$, end j of R_i is joined to end $\pi_i(j)$ of L_{i+1} (indices modulo t). Every vertex of the ring lies in one block, so a Petersen coloring of the ring is a labelling of the $3t$ joining edges accepted by every block: the ring is Petersen colorable if and only if the boolean matrix product $T_{M_1} \Pi_1 T_{M_2} \Pi_2 \cdots T_{M_t} \Pi_t$ has a nonzero diagonal entry, where Π_i is the permutation matrix of π_i .

Proposition 6.2. *For $q \in Q$ let S_q be the set of ordered label triples of class q .*

- (a) $T_M \subseteq \bigcup_{q \in Q} S_q \times S_q$, and every Π_i maps each S_q onto itself, so every product of transfer relations and junctions is block diagonal over Q .
- (b) *The diagonal blocks of such a product over two classes of one orbit are conjugate, so they have the same trace.*

Hence a ring is Petersen colorable if and only if, for some orbit, the product of the restrictions to $S_q \times S_q$, for one fixed class q of that orbit, has a nonzero diagonal. These six sector matrices have orders 60, 67, 48, 48, 30, 60.

Proof. (a) is the conduction property, and the class of a triple does not depend on the order of its entries. (b) An automorphism α of P maps a map σ of M to the map $\alpha \circ \sigma$, so $(x, y) \in T_M$ if and only if $(\alpha x, \alpha y) \in T_M$; α acts on labels and Π_i on positions, so they commute, and α conjugates the block over q to the block over $\alpha(q)$. The sector orders add up to $60 + 15 \cdot 67 + 15 \cdot 48 + 30 \cdot 48 + 30 + 2 \cdot 60 = 15^3$. $\square$

Corollary 6.3. *Let $\mathcal{F}$ be a finite set of blocks and $\mathcal{G}$ the set of six-tuples of sector matrices of $T_M \Pi$, for $M \in \mathcal{F}$ in either orientation and π any bijection. Every ring of at least two blocks of $\mathcal{F}$ is Petersen colorable if and only if every product of at least two elements of $\mathcal{G}$ has a nonzero diagonal in some sector. This holds if there is a set $\mathcal{C}$ of six-tuples of boolean matrices such that every product of two elements of $\mathcal{G}$ contains an element of $\mathcal{C}$ entrywise, every product cg with $c \in \mathcal{C}$ and $g \in \mathcal{G}$ contains an element of $\mathcal{C}$, and every element of $\mathcal{C}$ has a nonzero diagonal in some sector; and such a set exists whenever the condition holds (take all products).*

Proof. The first statement is Proposition 6.2. For the second, boolean products are monotone under entrywise containment, so by induction on the length every product of at least two elements of $\mathcal{G}$ contains an element of $\mathcal{C}$, and its diagonal is nonzero where that element's is. The set of all products is finite, which gives the converse. $\square$

A set $\mathcal{C}$ as in Corollary 6.3 is a certificate that anyone can check by products of boolean matrices of order at most 67. Entries of computed sector matrices are maps accepted by the checker, so a computed matrix can only miss entries; missing entries can only lower diagonals, and a certificate built from computed matrices is a certificate for the exact ones.

Claws and Petersen superedges. The *claw* Y is the block with a center joined to three vertices, each of which carries one dangling end in L and one in R . The *Petersen superedge* S is $P - u - w$ with u, w at distance 2, L the three ends at u and R the three ends at w ; it is the superedge $(P_{10})_{u,v}$ of Sedlar and Škrekovski [8]. The rings of claws contain the flower snarks of Isaacs [7]: J_k for odd k is the ring of k claws with the identity at $k - 1$ junctions and a transposition at the last one.

Theorem 6.4 (machine-verified). *Every ring of at least two blocks, each a claw or a Petersen superedge in either orientation, joined through arbitrary bijections, has a Petersen coloring. A ring of Petersen superedges always has one whose joining edges carry classes of the orbit E.*

Proof. The sector matrices of Y and S were computed by enumerating the models of a SAT formula projected to the dangling ends, each model accepted by the checker, and recomputed by exhaustive enumeration of maps and by a second encoding with a second solver, with identical results. The set of all products of at least two generators was closed exhaustively for $\{Y\}$, $\{S\}$ and $\{Y, S\}$: it has 30, 19,005 and 116,463 elements, all with a nonzero diagonal; for $\{S\}$ every element has a nonzero diagonal in the sector E. Certificates in the sense of Corollary 6.3, kept minimal under containment, have 12, 276 and 2,454 elements and were re-verified by a separate program with a different matrix product. $\square$

Corollary 6.5 (flower snarks). *For every odd $k \geq 3$, every Petersen coloring of J_k maps the three edges joining two consecutive claws onto an antipodal triple of P .*

Proof. The products $(Y \text{id})^{k-1}(Y \tau)$ are periodic in k from $k = 5$ with period 2, and for odd k only the sector T_1 has a nonzero diagonal (for even k only E). By conduction all junctions of a coloring carry one class, which is therefore T_1 , and the 30 ordered triples of class T_1 are the orderings of the five antipodal triples. $\square$

Hägglund and Steffen showed that flower snarks have Petersen colorings [6]; Corollary 6.5 describes all of them at the junctions.

Proposition 6.6. *Let S^t be the ring of t Petersen superedges with the identity at every junction.*

- (a) *For $t \geq 3$, S^t is a simple cubic graph, and it is 3-edge-colorable if and only if t is even.*
- (b) *For $t \geq 5$, S^t has girth 5 and is cyclically 5-edge-connected.*

So for every odd $t \geq 5$, S^t is a cyclically 5-edge-connected snark of girth 5 on $8t$ vertices, and by Theorem 6.4 it has a Petersen coloring.

Proof. (a) The same reduction applies to 3-edge-colorings, with the 27 color triples of a connector as states: the boundary relation of S , enumerated exhaustively, has powers of period 2, and the odd powers have a zero diagonal. (b) An exhaustive search finds no cycle-separating cut with at most four edges in S^5 , whose girth is 5. $P - u - w$ is connected, and restoring u and w gives P , which is 3-edge-connected, so a ring carries six edge-disjoint paths between any two blocks. The argument of Lemma 4.3, with cuts of at most four edges, at most four split blocks and runs of at most four consecutive blocks (all of which occur in S^5), shows that S^t has no cycle-separating cut with at most four edges for $t \geq 5$, and a cycle of length at most four would lie in such a run. $\square$

Remark 6.7 (the classes are not enough). G_{52} has a 6-edge cut with sides of 18 and 34 vertices. For each of the ten splits of the cut into two triples, the two sides are 6-poles with Petersen colorings whose conducted sets share 0, E and D_2 , so no argument by classes separates them; their ring of length 2 is G_{52} , and the product of their sector matrices has a zero diagonal in all six sectors, as it must. In a further test, 203 rings (the flower snarks J_5 , J_7 , J_9 and 200 random words of length 2 to 6 over 81 blocks, with 18 to 132 vertices) were decided both by the sector matrices and by a SAT solver on the graph, with the same answer every time.

7 Remarks and questions

Other 4-poles. The computation behind Proposition 5.1 was carried out for a wider set of 4-poles (computation EXP-012). The Petersen graph minus two adjacent vertices, and the Petersen graph minus two edges at distance 2 or 3, transmit distance 1 under every pairing of their dangling ends. For G_{52} minus two adjacent vertices, with the pairing of Lemma 4.2(b), the transmitted set is $\{0\}$ for eleven of the fourteen edge orbits of the automorphism group of G_{52} , $\{0, 3\}$ for two and $\{0, 2\}$ for one; it never contains 1, as Lemma 4.2(b) requires. Under the two other pairings every such 4-pole transmits distance 1. Among the 4-poles $G_{52} - ab - cd$ with the connectors of Lemma 4.2(a), 282 of the 546 edge pairs examined have $D = \{1\}$ (computation EXP-010). Many rings with the property of Theorem 4.1 can therefore be built from G_{52} alone.

Criticality. A pair of vertices of a graph without a Petersen coloring is *critical* if some map has exactly these two vertices bad. In every one of fifteen cyclically 4-edge-connected counterexamples examined in [10] every pair of adjacent vertices is critical. By Theorem 4.1 the two bad vertices of a map of R_2 lie in two non-consecutive blocks, so two adjacent vertices of one block of R_2 form a non-critical pair, and in R_t with $t \geq 3$ no pair is critical. Universal 2-criticality is therefore not a property of the cyclically 4-edge-connected class.

Questions. The rings have 4-edge cuts, and no cyclically 5-edge-connected counterexample to the Petersen coloring conjecture is known [5, Problem 11]. A ring of 6-poles without a Petersen coloring needs blocks whose semigroup in Corollary 6.3 contains a product with a zero diagonal in every sector; which 6-poles without small internal cuts have this property? Is the Petersen defect bounded on cyclically 5-edge-connected cubic graphs? What is the largest constant c such that cubic graphs with $\text{pd} \geq cn$ exist in the cyclically 4-edge-connected class? The rings give $c \geq 1/102$.

Data availability

The edge list of G_{52} , the encoders, the checkers, the ring construction, the cut searches and all results, with the SHA-256 digests of every formula and proof, are in the repository https://github.com/fsantibanezleal/CAOS_RESEARCH, under `problems/combinatorics/petersen-coloring/`. The computations of this paper are archived in the folder `experiments/EXP-012-conduction-rings` of that directory (runners `run.py` and `run_ring.py`; library modules `code/pcplib/poles.py` and `code/pcplib/rings.py`), with the distance sets of [10] in `experiments/EXP-010-sublinear-approximation`. The upper bound of Proposition 5.3 is in the same folder (`run_r3_upper.py`). Section 6 is archived in `experiments/EXP-013-six-pole-charges` (conducted sets) and `experiments/EXP-014-transfer-semigroups` (sector matrices, closures, certificates and their checker `check_certificate.py`, the control on G_{52}), with the library modules `code/pcplib/charges.py` and `code/pcplib/transfer.py` and the probes `code/probes/superedge_ring_*.py`; the theory notes are in `context/2026-10-06-charges.md`. Proofs were produced by CaDiCaL [11] and checked with `drat-trim` [12].

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